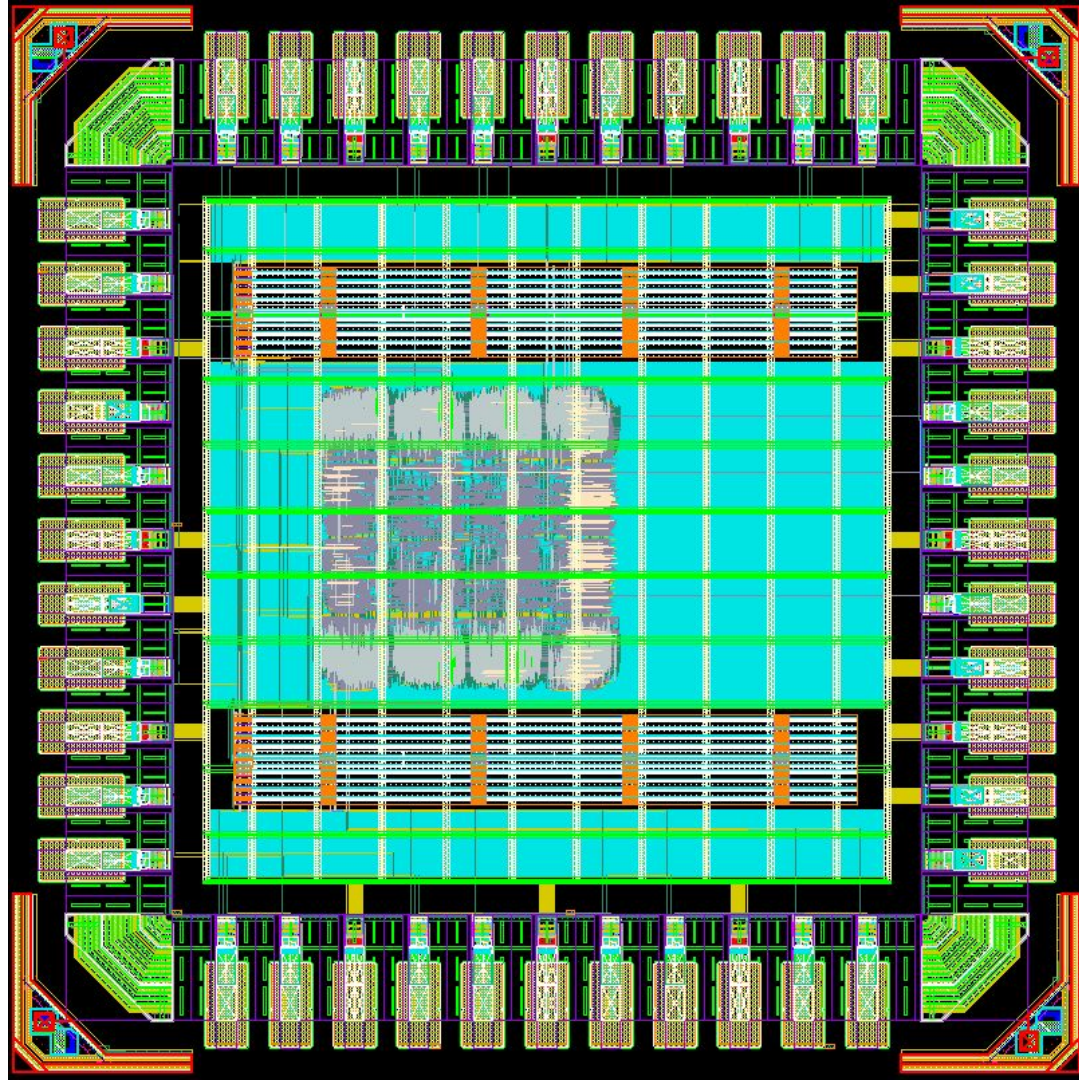


(FPGA--)-- A Retrospective

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Layout



Timeline of Design

1. DR1: 10x10 fabric, hand layout
2. ADR1: 30x30 fabric, full custom
3. DR2: 20x20 fabric, synthesized
4. ADR2: 4x4 fabric, synthesized, simplified
5. Tapeout: Same as ADR2, extra I/O mode

Not how you want to design a chip --

But we did meet tapeout!

External Toolchains

- Cause of a lot of grief and blocking -- but not necessarily a bad thing!
 - Testing blocked on fully working VTR flow
- Toolchain needs to be considered in the design
 - Have a backup plan ready!
- Set hard deadlines to switch to backup plan
 - Resolve the block

Pivoting

- Major re-architecting and re-implementation after spring break
 - Only two blocks (CBs and SBs) from the old design are used in the tapeout
- Painful and scary, but necessary
 - Discarded lots of prior effort (painful)
 - Starting almost from scratch halfway through the semester (scary)
 - Eliminated blocking issues (necessary)

Alternative Flows (bottom-up)

- A regular design is very amenable to bottom-up
- But goes against the tools sensibilities
- Taking a lot of matters into our own hands
 - Know exactly what we want
 - Tell the tool what exactly we want
 - Verify that it gets us

Scope/Previous Preparation/Words of Wisdom

- If you don't have RTL in hand at the start, the design had better be simple
- Build RTL/infrastructure to be scalable
 - Might have a lot less area/Tclk/development time than you thought
- Communication is key
 - Plan to avoid serializing on one person's tasks

PAR, LVS and DRC Fun

- Do we know what we want? No! Do we know what we can do? No!
 - Course Staff to the rescue
- LVS errors are the least logical
 - Where is the error!?
- DRC: after everything... just a chore



99 little bugs in the code.
99 little bugs in the code.
Take one down, patch it around.
127 little bugs in the code...